

Design and development Matrix

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Needs analysis		Target specifications			Technical solution			Verification			Validation			Complete
Ref/Need	Description	Ref Spec	Description	Objectives measurable + tolerance	Possible solutions	The chosen one	References	Verification test description	Location/Team	Control methodology	Validation test description	Location/Team	Control methodology	Y/N
N1	PERFORMANCE - Reduce lap time by minimising friction between the robot and track surface during motion.	SP1	Lap time performance	Complete lap in 20s +/- 5s	• 4-wheel chassis (higher stability, higher friction) wheel + skid design • Bail caster wheel design	• 2-wheel drive with rear low-friction skid. Selected based on speed (40%), friction reduction (30%), stability (20%), simplicity (10%). Reduces contact area so directly improves lap time target	[1]	Time full lap using stopwatch over 5 trials and calculate average lap time	Group 18	Use fully charged battery for all runs, same track layout, identical start position, and no external interference				
N2	PERFORMANCE - Accurately detect the electromagnetic signal from the track wire without distortion.	SP2	Signal detection accuracy	Detect signal with > 90% reliability across during testing using testing tracks	• Simple diode rectifier • Precision rectifier circuit • Direct AC signal processing	Precision rectifier + peak detector. Chosen based on accuracy (45%), signal clarity (25%), noise reduction (20%), complexity (10%). Meets detection reliability requirement better than simple rectifier.	[2]	Compare detected signal vs known track signal using oscilloscope and count correct detections	Group 18	Reliability percentage calculation (>90%), multiple track positions tested				
N3	PERFORMANCE - Deliver sufficient power to motors without excessive energy loss during operation.	SP3	Motor efficiency and control	Maintain speed with < 10% drop over 5 runs	• Linear transistor motor driver • MOSFET switching driver • Direct battery connection	MOSFET switching motor driver. Chosen based on efficiency (40%), thermal performance (25%), speed control (20%), implementation (15%). Minimises power loss vs linear driver.	[3]	Measure speed consistency over 5 runs using tachometer or lap timing	Group 18	% speed drop calculation (<10%), controlled battery conditions				
N4	PERFORMANCE - Achieve smooth and controllable motor speed to prevent sudden acceleration and loss of track.	SP4	Motor control smoothness	Use tachometer to analyse rpm. Maintain stable motion with no sudden speed spikes and < 10% variation in speed during turns.	• Fixed voltage control • PWM (pulse width modulation) control • Manual resistance-based speed control	PWM motor control. Chosen based on control precision (40%), efficiency (25%), responsiveness (20%), implementation complexity (15%). Provides smooth speed adjustment, directly improving control stability and track following.	[4]	Observe motor response to speed changes and turning behaviour during track run	Group 18	Qualitative smoothness + RPM variation measurement (<10%)				
N5	USABILITY - Allow quick repositioning of the sensing coil during testing to optimise wire detection.	SP5	Ease of adjustment	Adjustment completed in < 2 min without full rebuild	• Permanently soldered coil to PCB • Mechanical mount (screw/bracket) • Temporary fixing using sellotape + crocodile clips	Temporary mounting using sellotape and crocodile clip connections to the main circuit. Chosen based on speed of adjustment (40%), flexibility (30%), no permanent modification (20%), simplicity (10%). Enables quick repositioning during iterative testing.	[5]	Time how long it takes to reposition sensing coil during testing	Group 18	Stopwatch measurement (<2 min), repeat across multiple adjustments				
N6	BUSINESS - Minimise cost by selecting affordable and readily available components.	SP6	Cost efficiency	Total cost of the system to department < £30 +/- £5	• Custom PCB design • Off-the-shelf components • High-performance specialised parts	Off-the-shelf lab components. Chosen based on cost (40%), availability (30%), performance (20%), ease of integration (10%). Meets cost constraint.	[6]	Calculate total cost using BOM, and receipts from online sources	Group 18	Compare total cost against £30 +/- £5 constraint				
N7	HEALTH AND SAFETY - Prevent fumes from being inhaled during soldering of components onto stripboard.	SP7	Safe assembly practice	Maintain safe working distance > 30 cm from solder point and ensure visible fume displacement	• Wear respirator mask • Use extraction fan • Maintain safe distance from solder point	Use of local fan to direct fumes away from user. Chosen based on effectiveness (40%), practicality (30%), availability (20%), comfort (10%). Reduces inhalation risk during build.	[7]	Observe soldering process and confirm fumes are directed away from user in the direction of the fan	Group 18	Visual inspection + adherence to safe working distance (>30 cm)				
N8	SUSTAINABILITY - Reduce waste by reusing electrical components across multiple design iterations instead of discarding parts.	SP8	Material sustainability	> 80% components reused across builds	• Single-use soldered design • Fully modular reusable design • Hybrid reuse approach	Hybrid reuse approach (reuse key components, rebuild minor sections). Chosen based on sustainability (40%), practicality (30%), time efficiency (20%), performance retention (10%).	[8]	Record number of reused vs new components across builds	Group 18	Percentage calculation (>80% reused components)				
N9	AESTHETIC - Maintain a neat and organised wiring layout to improve readability of the connections and for professionalism.	SP9	Visual clarity	No loose wires > 3cm, clear routing visible	• Loose wiring • Cable-tied routing • Fully enclosed casing • No excessively long cut wires	Reduce and remove unnecessary wire length. Chosen based on clarity (35%), ease of debugging (30%), aesthetic quality (20%), accessibility (15%).	[9]	Visually inspect wiring layout and measure loose wire lengths using ruler. Assess clarity of routing and accessibility for debugging.	Group 18	Use same circuit setup and component layout across tests, assess under consistent lighting conditions, and apply identical measurement method for wire length.				
N10	SOCIETAL / INCLUSION - Ensure all team members can understand and operate the robot regardless of technical background.	SP10	Accessibility of operation	All members operate system after < 10 min explanation	• Complex undocumented system • Basic labelled system • Fully documented + labelled system on GitHub and Teams	Fully documented and labelled system. Chosen based on clarity (40%), ease of use (30%), team inclusivity (20%), time efficiency (10%).	[10]	Allow new user to operate robot after explanation and observe success	Group 18	Time-based evaluation (<10 min learning), task completion check				
N11	STANDARDS - Ensure the electrical system operates within safe limits aligned with recognised standards.	SP11	Electrical safety compliance	Operate at < 6 V DC, no exposed conductors	• No compliance consideration with standards • Basic safe design • Standards-informed design	Designed in line with BS EN 61010-1 (lab electrical safety), with consideration of RoHS Directive 2012/19/EU and WEEE Directive 2012/19/EU. Provides stable, compliance (30%), professional practice (20%), documentation (10%).	[11]	Inspect circuit for exposed conductors and measure operating voltage	Group 18	Voltage verification (<6V DC) + visual safety inspection				
N12	INCLUSION / DIVERSITY - Ensure all team members can contribute ideas during design discussions without exclusion or dominance by a single member.	SP12	Inclusive communication and teamwork	Each team member contributes at least 1 idea per meeting, with no single member contributing > 50% of total discussion input	• Unstructured discussion (free conversation) • Rotating speaking turns system • Designated team leader controlling discussion	• Rotating speaking turns during discussions. Chosen based on inclusivity (40%), fair participation (30%), idea diversity (20%), ease of implementation (10%). Ensures balanced contribution and aligns with inclusive teamwork objectives.	[12]	Record team meeting contributions from each member	Group 18	Contribution tracking (>1 idea per member, <50% dominance). Project Management lead in charge of keeping track.				
N13	ENVIRONMENTAL - Reduce plastic waste when fabricating the low-friction skid.	SP13	Material sustainability	Ensure > 80% of skid material is reused or repurposed, with minimal new plastic used (< 20%)	• Use a new acrylic plastic sheet • Reuse skids from previous year projects • Use scrap/offcut plastic from workshop stock	Use scrap/offcut plastic from workshop stock. Chosen based on waste reduction (45%), material availability (25%), environmental impact (20%), cost (10%). Minimises new material usage while maintaining adequate performance.	[13]	Track source of skid material (new vs reused). From a workpiece, calculate percentage used via weight of skid/weight of all material.	Group 18	Percentage reused (>80%), estimate new material usage (<20%)				
N14	HEALTH AND SAFETY - Safely manufacture the low-friction skids using appropriate tools and methods in the workshop.	SP14	Safe fabrication practice	Complete fabrication with 0 injuries/incidents and maintain controlled cutting (no uncontrolled fractures or slips)	• Use clamp and saws for controlled cutting • Bend plastic and snap along scored line • Use laser cutter	• Blue clamp and saws for controlled cutting. Chosen based on safety (40%), control/precision (30%), availability (20%), material suitability (10%). Provides stable, controlled cutting and reduces risk of sudden breakage compared to snapping, while being more accessible than a laser cutter.	[14]	Monitor fabrication process and record any incidents	Group 18	Zero incident requirement + controlled cutting verification. Adherence to workshop etiquette and rules.				
N15	PERFORMANCE, USABILITY, SUSTAINABILITY - Reduce overall mouse weight to improve overall efficiency and ease of handling during testing.	SP15	Energy efficiency and ease of use	Total robot mass < 500 g +/- 50 g, with consistent performance (< 10% speed drop over 5 runs)	• Compact and tightly integrated design • Use lighter skid material • Wirelessly power the system	• Compact and tightly integrated design. Chosen based on weight reduction (40%), practicality (30%), reliability (20%), implementation feasibility (10%). Reduces unnecessary mass without introducing complexity or instability from wireless power.	[15]	Measure total mouse mass using scale and test performance over 5 runs	Group 18	Mass check (<500 g +/- 50 g) + performance consistency (<10% drop). Ensure scale unit is consistent over rounds and ensure no zero-error.				

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